

□ Topic of the session: Advanced Java Programming

□ UNIT 8: **SERVLETS**

□ **University of Kigali (UoK)**

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SERVLETS



Servlets

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What is Servlet?

Java Servlets are programs that run on a Web or Application server and act as a middle layer between a request coming from a Web browser or other HTTP client and databases or applications on the HTTP server.

What is Tomcat?

It is an open-source Java servlet container that implements many Java Enterprise Specs such as the Websites API, Java-Server Pages (JSP) and last but not least, the Java Servlet.

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In simple words, we can say that the **Apache Tomcat is actually a server and a servlet container.**

Servlet

We will use the following tool:

- **Eclipse 4.6 (NEON)**
- **Servlet 3.0**
- **Tomcat 8**

Using Servlets, you can collect input from users through web page forms, present records from a database or another source, and create web pages dynamically.

8.1 Install Tomcat Web Server for Eclipse

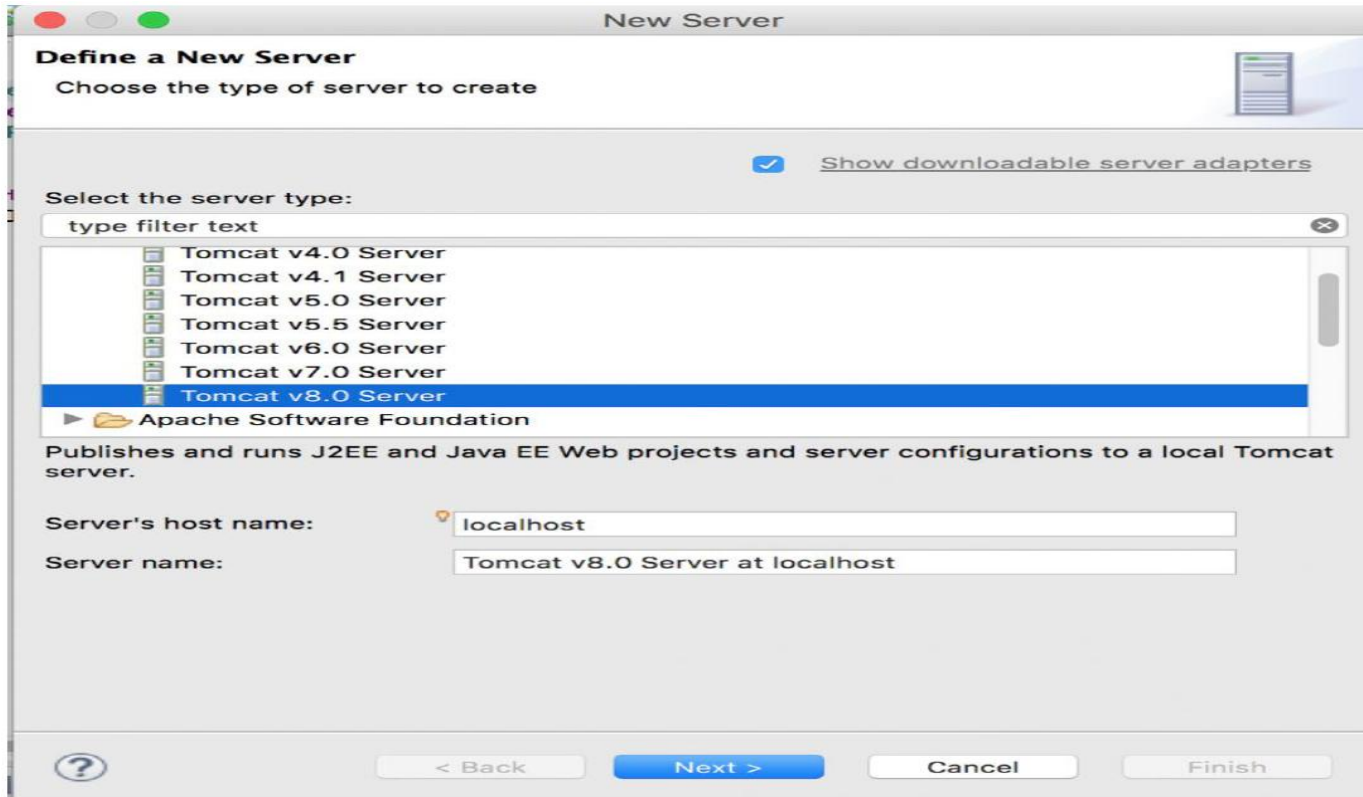
To begin with **Servlet**, you need to download **Tomcat Web Server** and declare it with **Eclipse**.

Go to this link: <http://tomcat.apache.org/download-80.cgi>, and Extract the zipped folder to any desired location. **Select the Server in Eclipse:** Open Eclipse IDE › Click on the Servers tab located at the bottom › right click › New › click on Server.

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8.1 Install Tomcat Web Server for Eclipse

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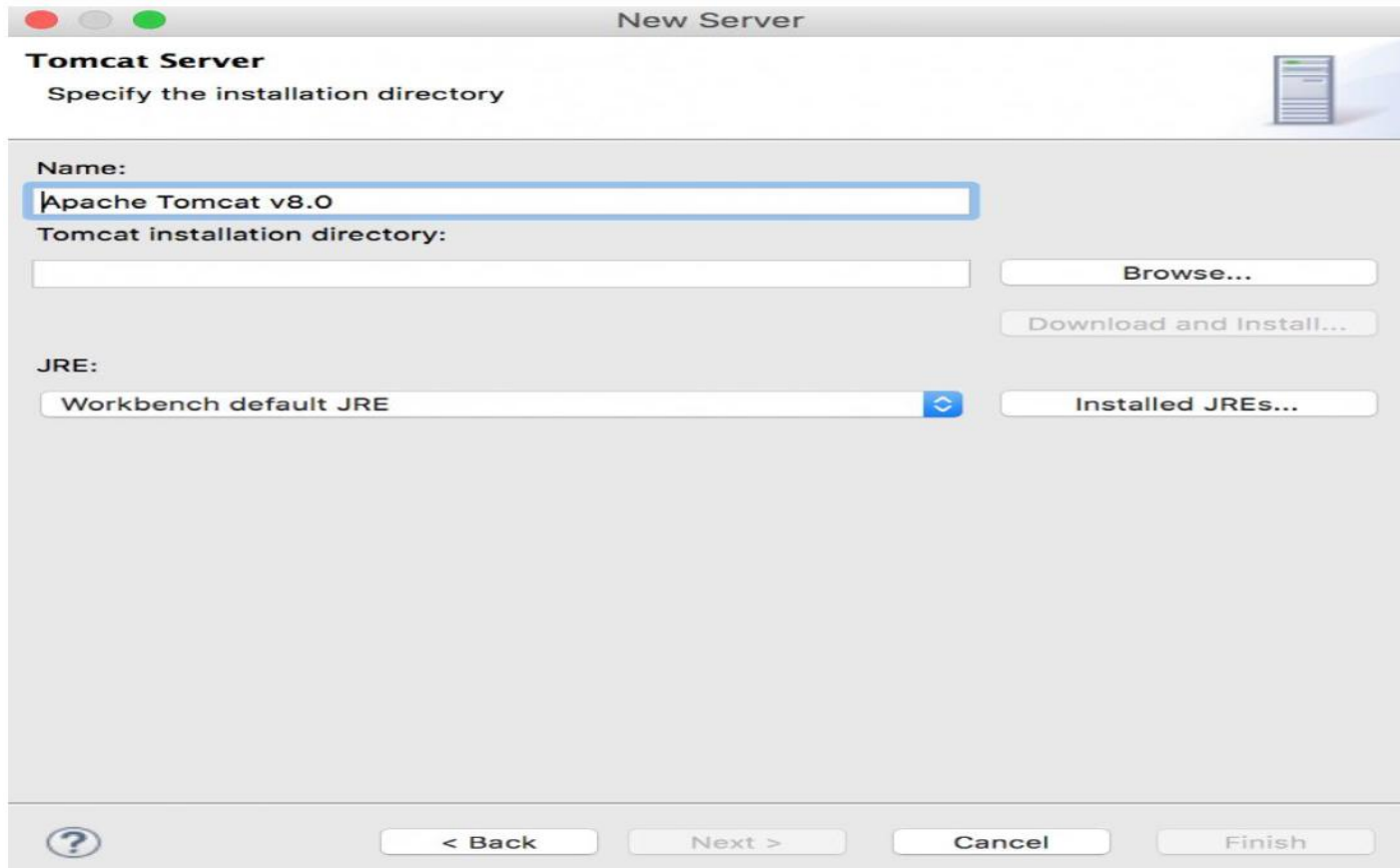


You will be presented with a window as shown in the image below. Click browse (next slide) and select the folder that you have extracted from the zip file in Step 2.

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8.1 Install Tomcat Web Server for Eclipse

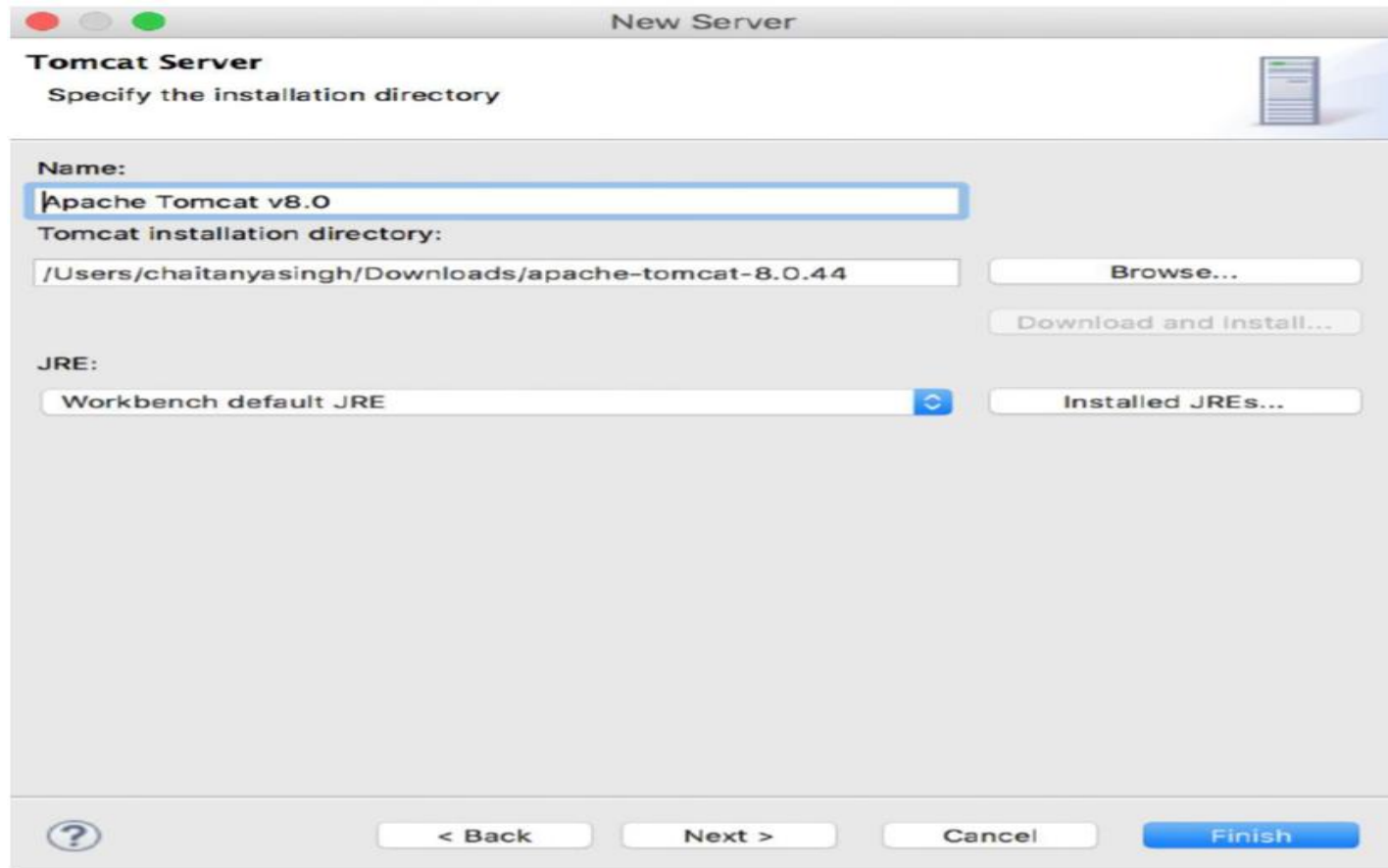


Click finish.

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8.1. Install Tomcat Web Server for Eclipse



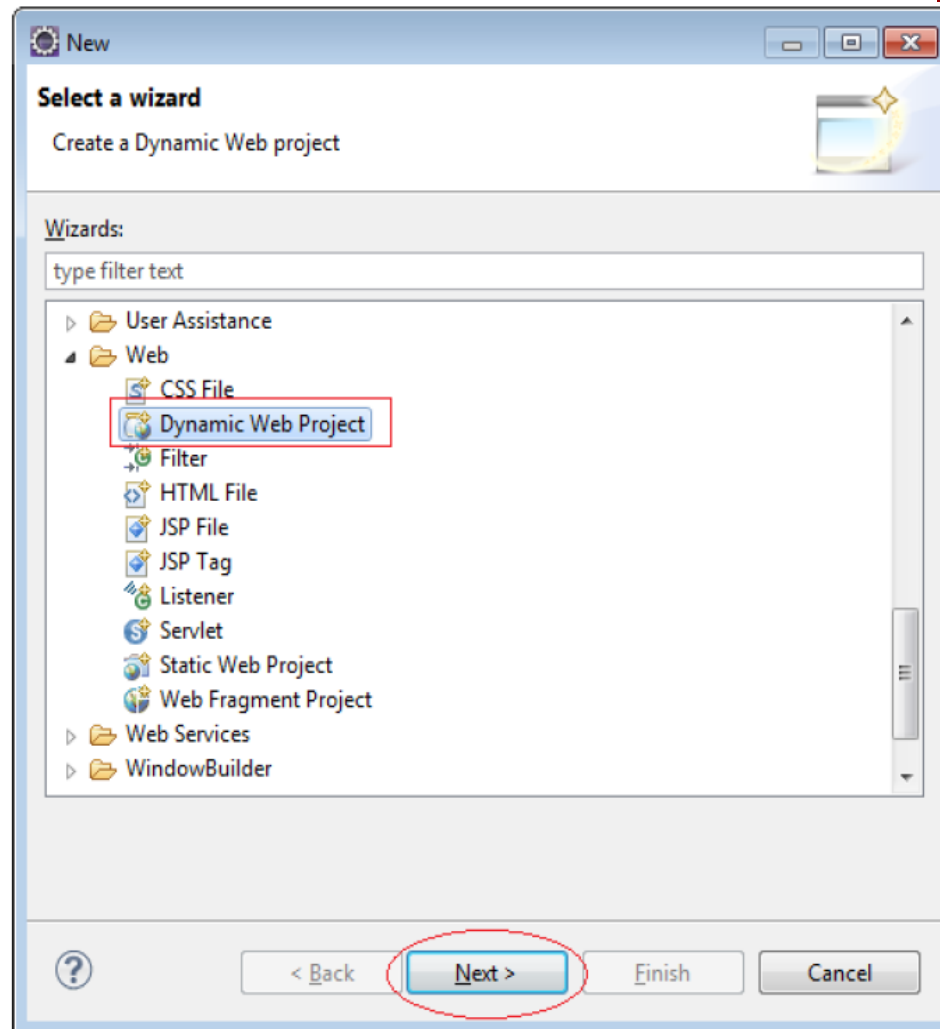
You have successfully configured the tomcat server in Eclipse, you can now run the Servlet and JSP (Java Server Pages) in Eclipse.

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8.2 Create Web Project to start with Servlet

➤ File/New/Other



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8.2. Create Web Project to start with Servlet

Project Name: ServletTutorial

New Dynamic Web Project

Create a standalone Dynamic Web project or add it to a new or existing Enterprise Application.

Project name:

Project location

☒ Use default location

Location:

Target runtime

Dynamic web module version

Configuration

The default configuration provides a good starting point. Additional facets can later be installed to add new functionality to the project.

EAR membership

☐ Add project to an EAR

EAR project name:

Working sets

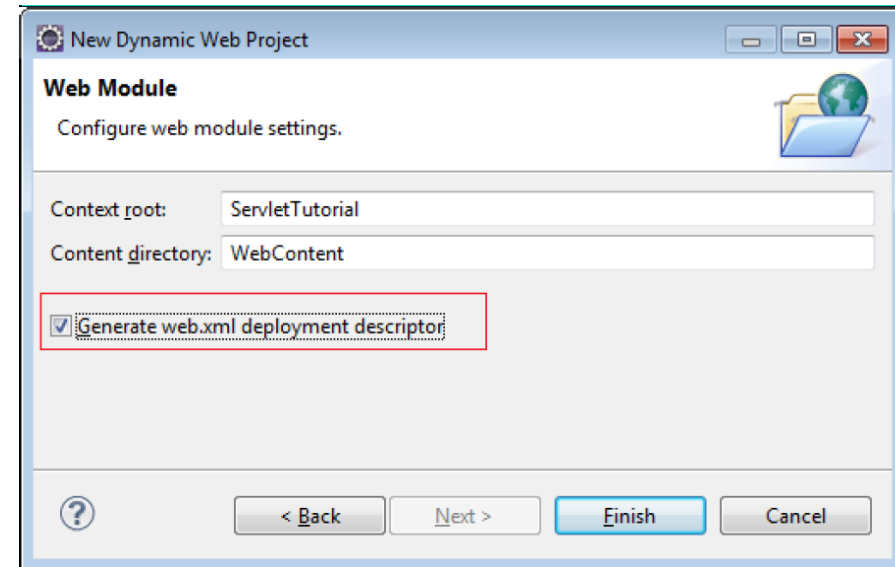
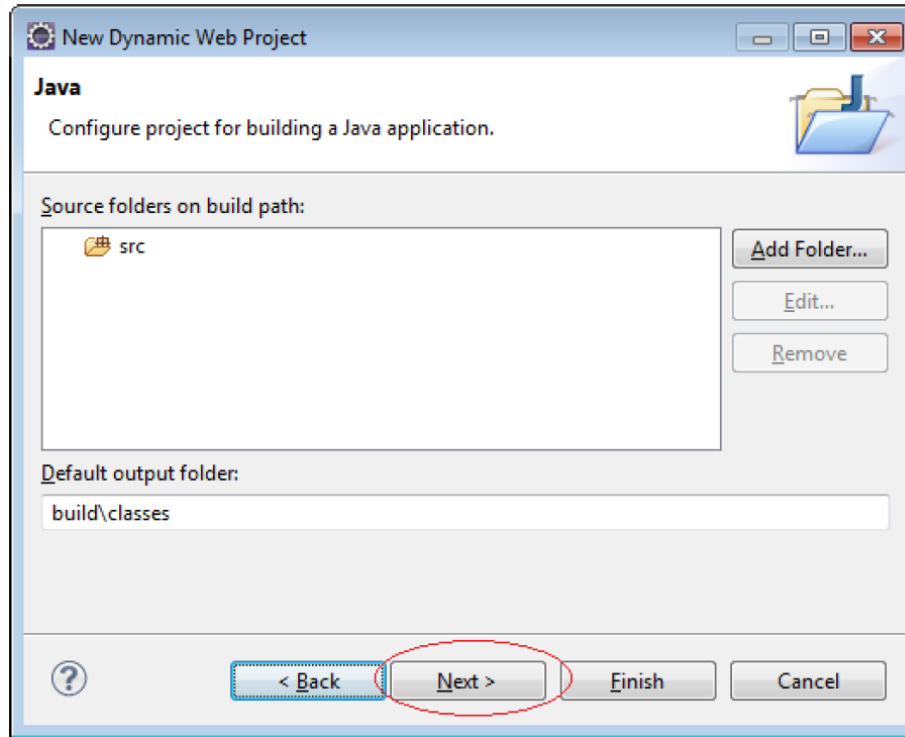
☐ Add project to working sets

Next

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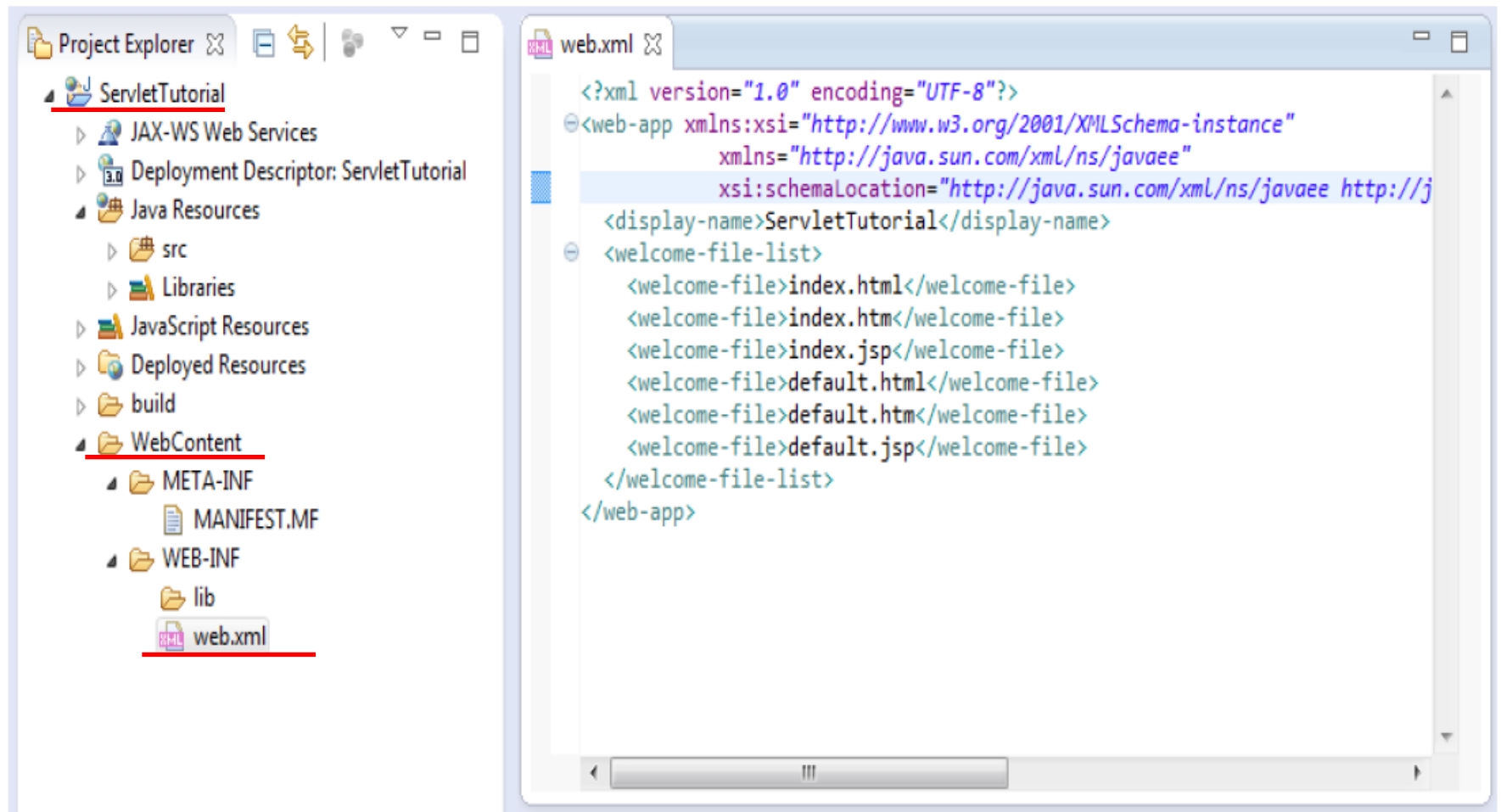
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8.2. Create Web Project to start with Servlet

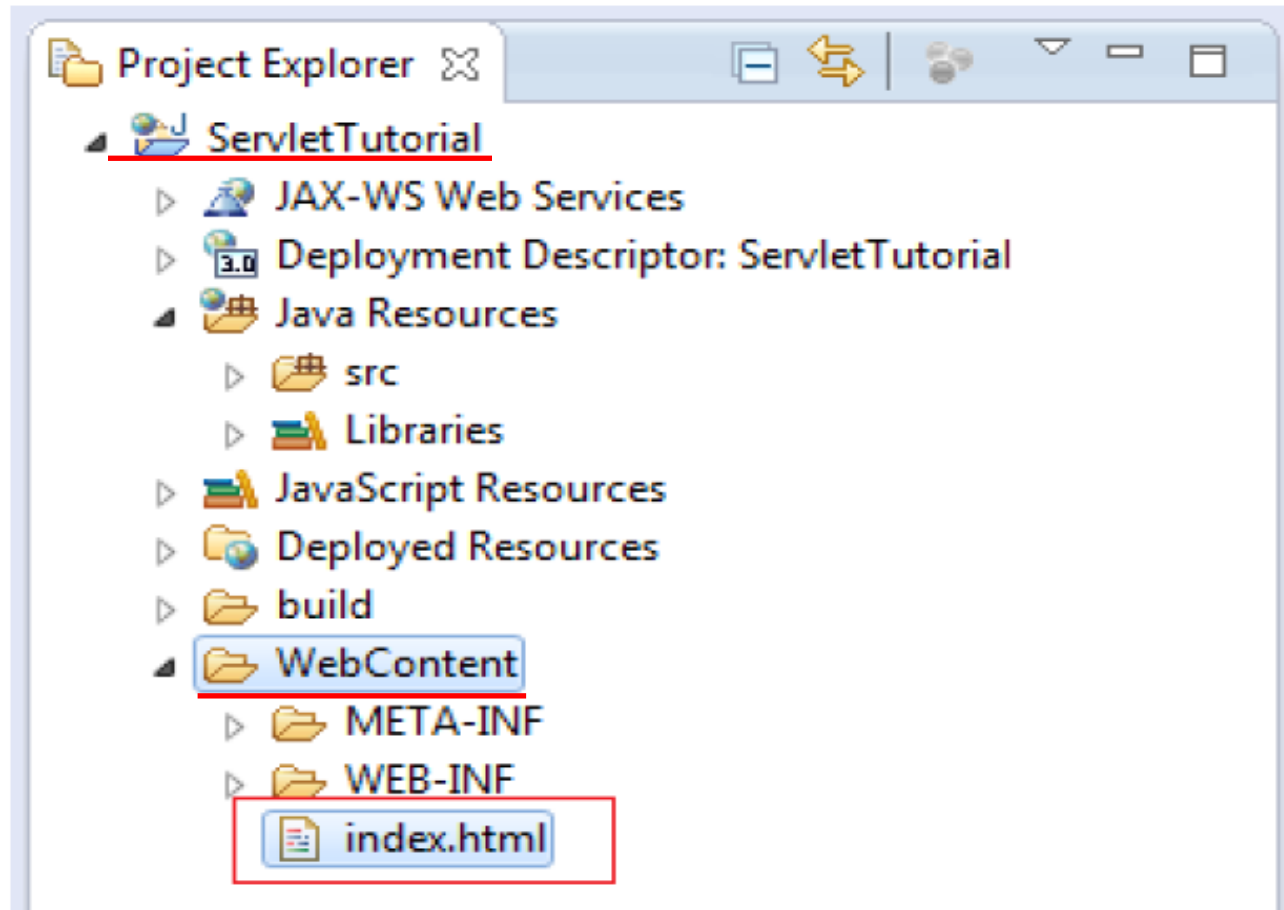


Project was created:

8.2. Create Web Project to start with Servlet



8.2. Create Web Project to start with Servlet



Create file *index.html*:

8.2. Create Web Project to start with Servlet

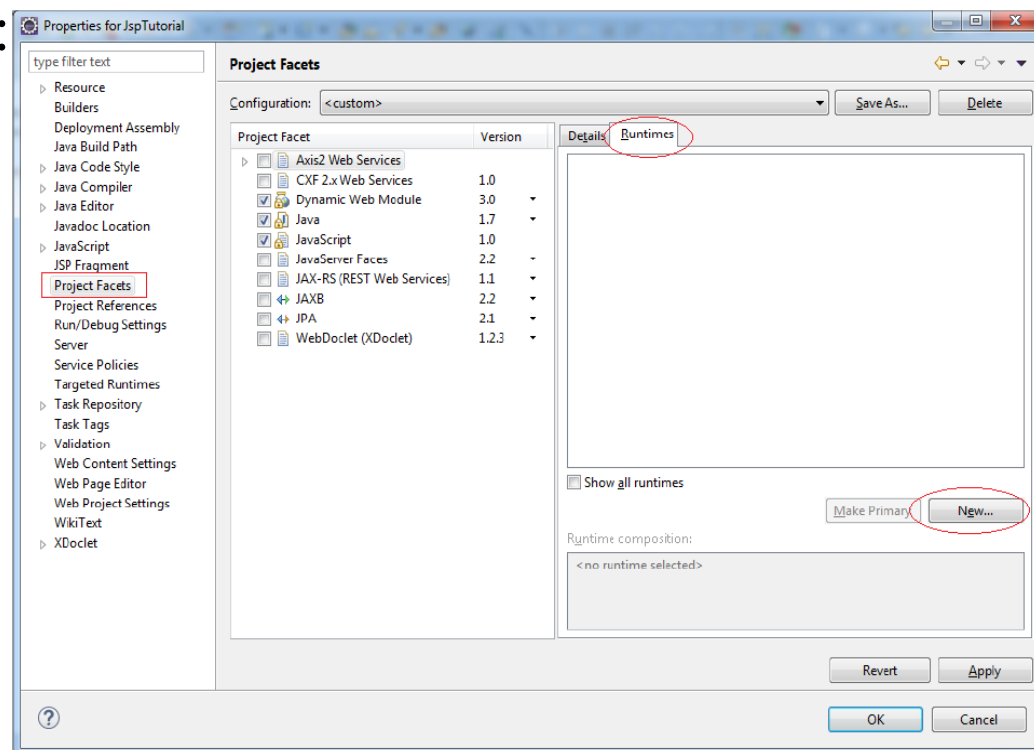
index.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
<h1>Hello World</h1>
</body>
</html>
```

8.3. Configure Eclipse to run apps on tomcat

So far, we have not yet done anything with **Servlet**. Now we configure it so that it can run that web application. Then we start with **Servlet**.

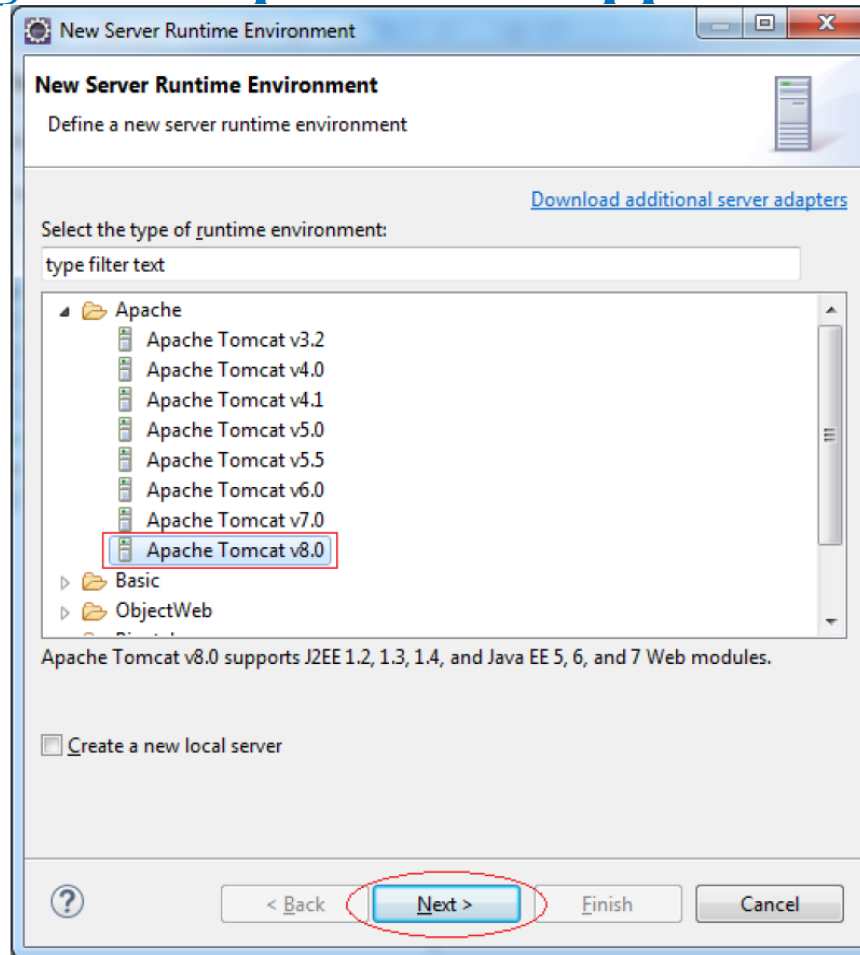
In eclipse, **right-click** on the **Project ServletTutorial** (project name), select **Properties**:



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8.3.. Configure Eclipse to run apps on tomcat

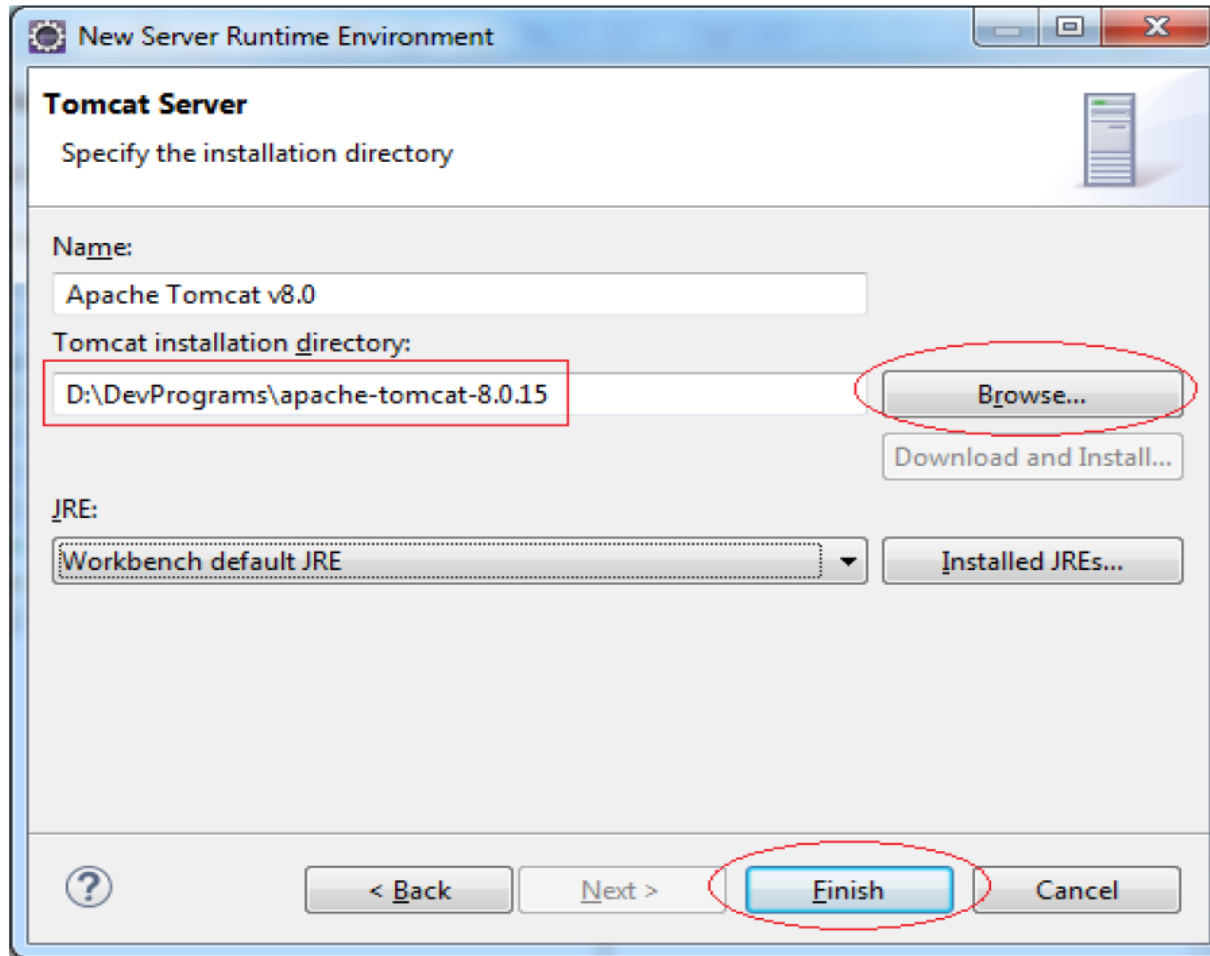


Select the location of your Tomcat.

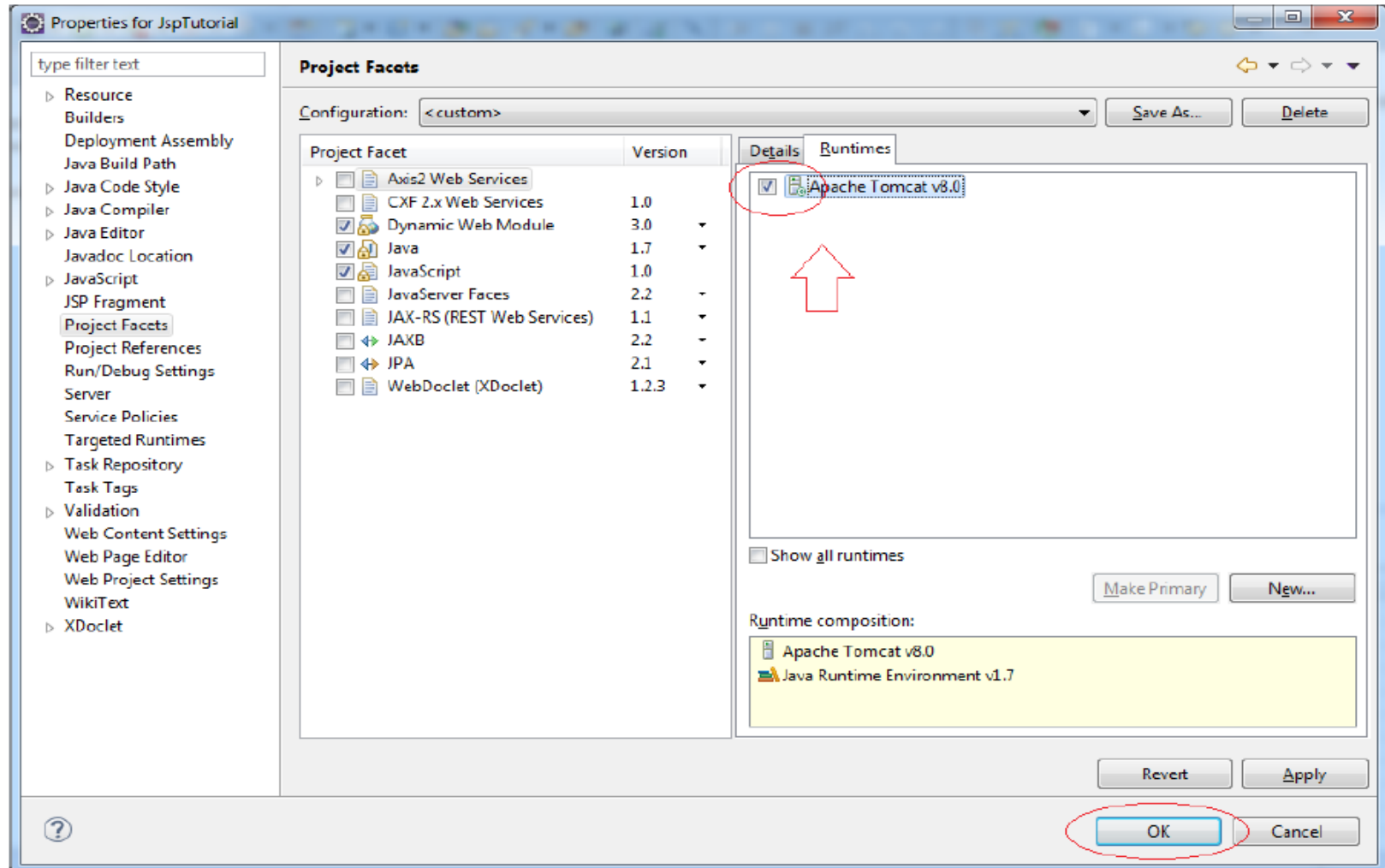
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8.3. Configure Eclipse to run apps on tomcat



8.3. Configure Eclipse to run apps on tomcat

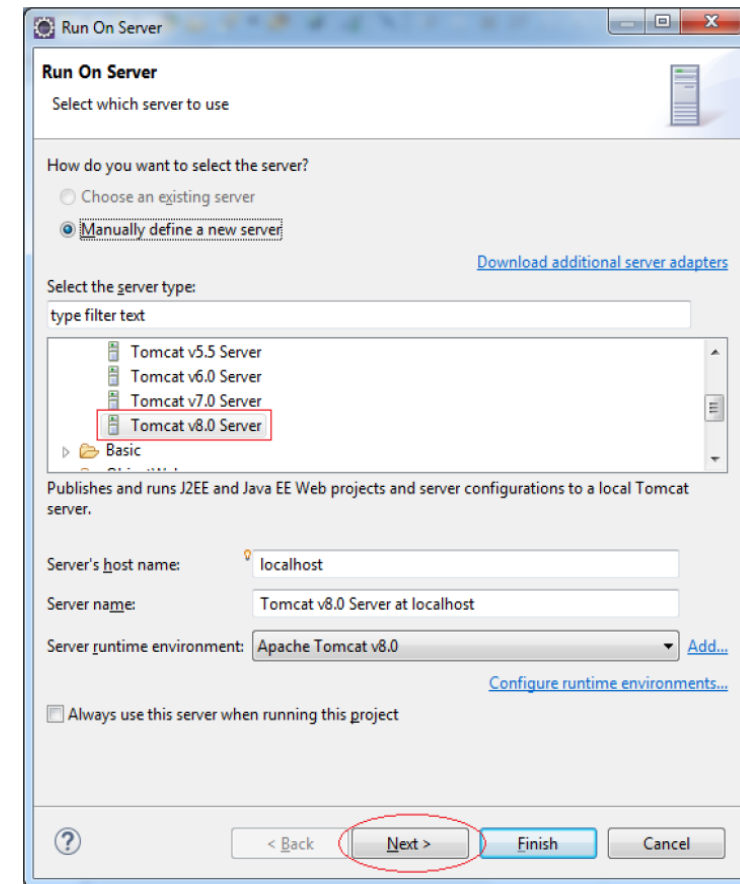
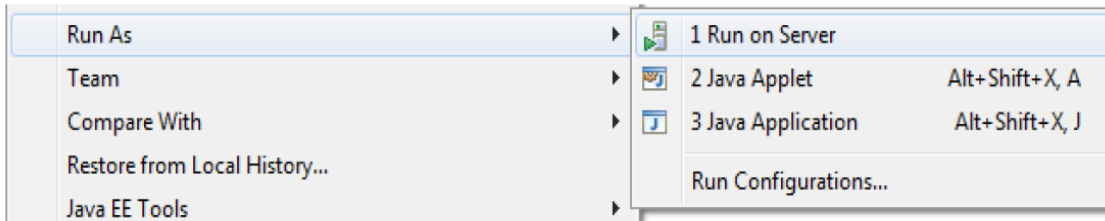


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8.3. Configure Eclipse to run apps on tomcat

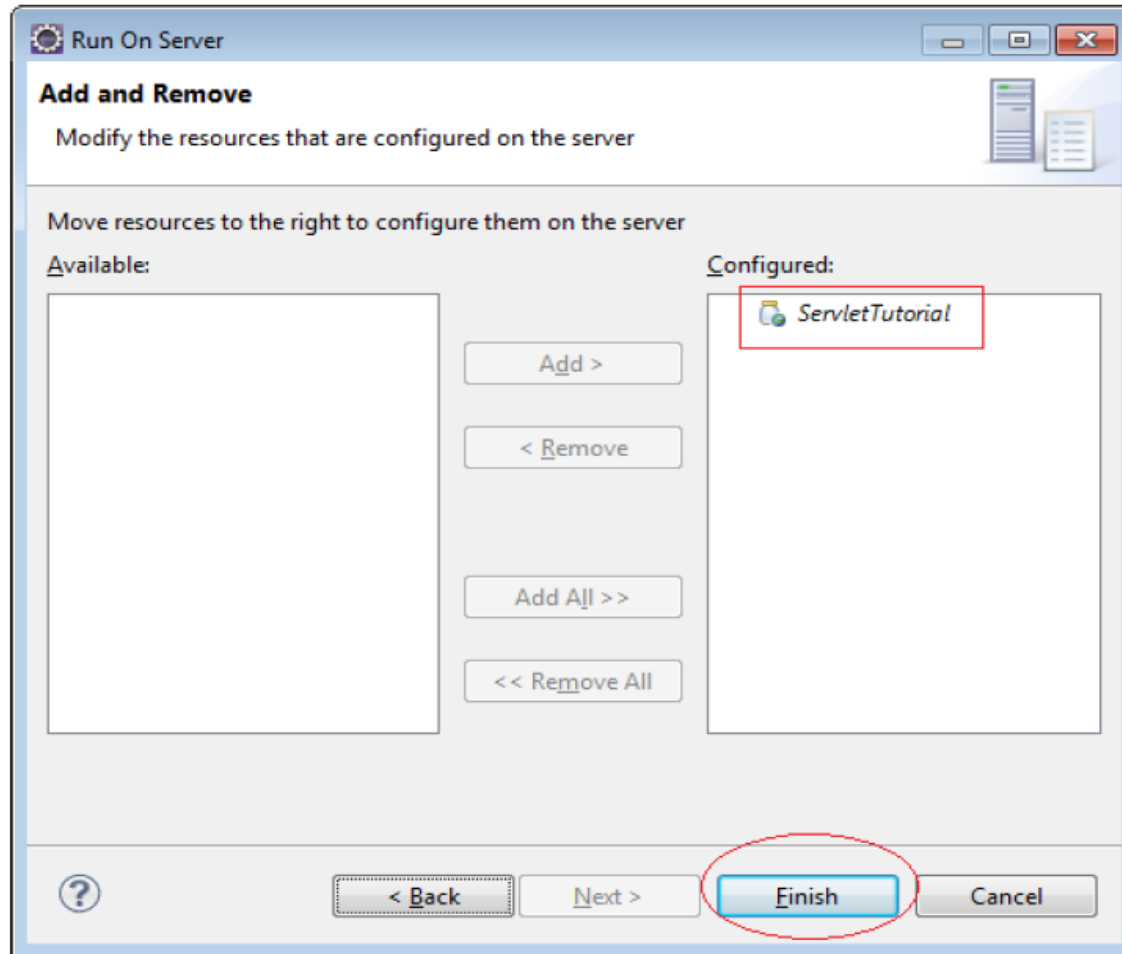
Right-click the project **ServletTutorial**, select **"Run As / Run on Server"**



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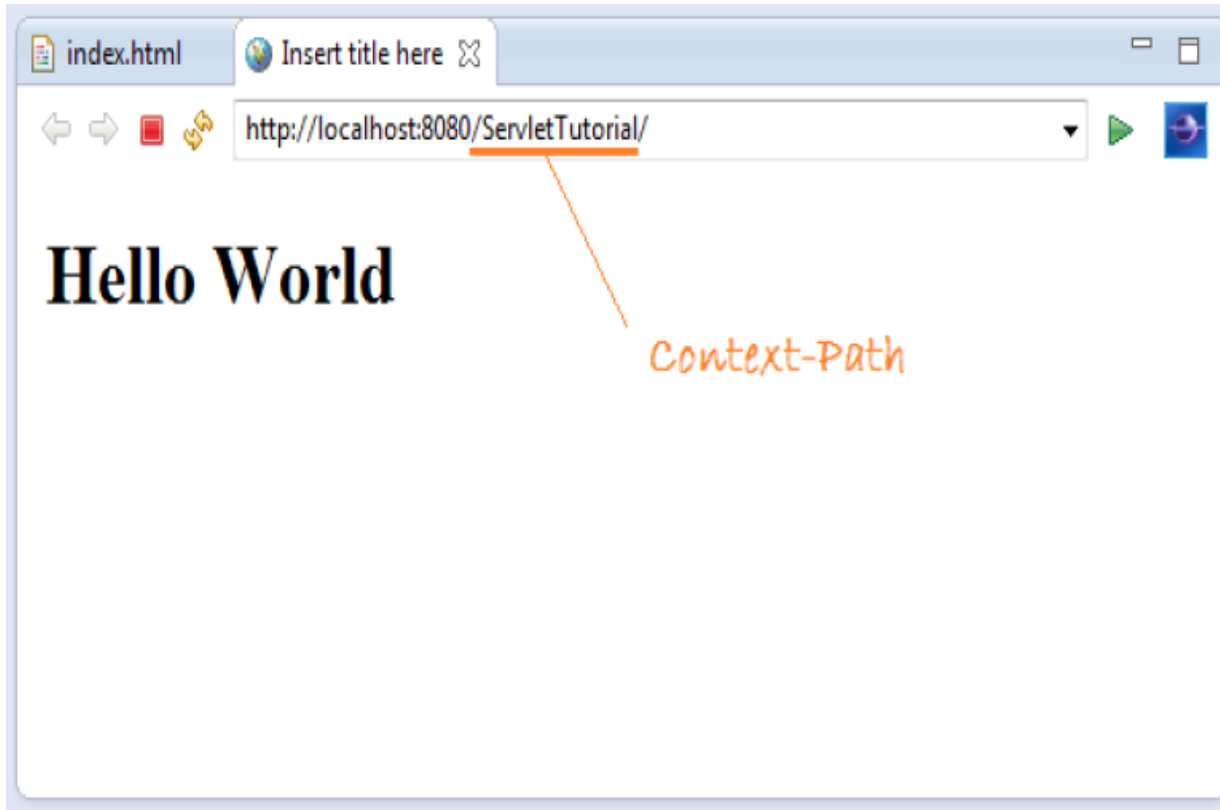
8.3. Configure Eclipse to run apps on tomcat



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8.3. Configure Eclipse to run apps on tomcat



Website is running on Eclipse browser.

HelloServlet.java example:

```
package josbert_new_;

import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.ServletOutputStream;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class HelloServlet extends HttpServlet {

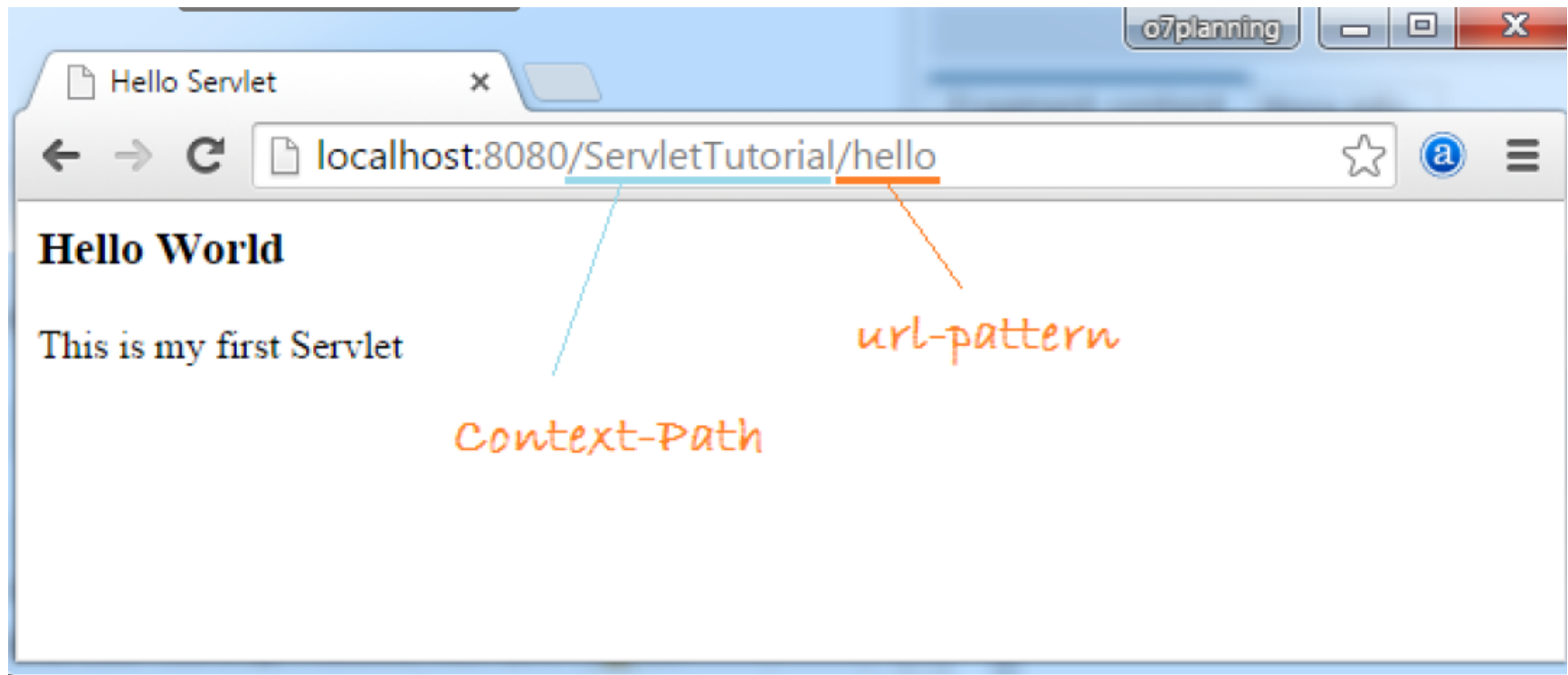
    private static final long serialVersionUID = 1L;
    public HelloServlet() {
    }
    @Override
    protected void doGet(HttpServletRequest request,
        HttpServletResponse response) throws ServletException, IOException {
        ServletOutputStream out = response.getOutputStream();
        out.println("<html>");
        out.println("<head><title>Hello Servlet</title></head>");
        out.println("<body>");
        out.println("<h3>Hello World</h3>");
        out.println("This is my first Servlet");
        out.println("</body>");
        out.println("</html>");
    }
    @Override
    protected void doPost(HttpServletRequest request,
        HttpServletResponse response) throws ServletException, IOException {
        this.doGet(request, response);
    }
}
```

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HelloServlet.java example:

Sample output:



Add/Product of Numbers.java example:

```
package josbert_new_;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.http.*;
public class add_numbers_cmts extends HttpServlet {

    // The 'service' method handles requests from the browser (GET or POST)
    public void service(HttpServletRequest request, HttpServletResponse response) throws IOException {

        // Get the value of 'num1' from the request and convert it from
        String to int
        int num1 = Integer.parseInt(request.getParameter("num1"));

        // Get the value of 'num2' from the request and convert it from
        String to int
        int num2 = Integer.parseInt(request.getParameter("num2"));

        // Calculate the sum of the two numbers
        int sum = num1 + num2;

        // Calculate the product (multiplication) of the two numbers
        int product = num1 * num2;

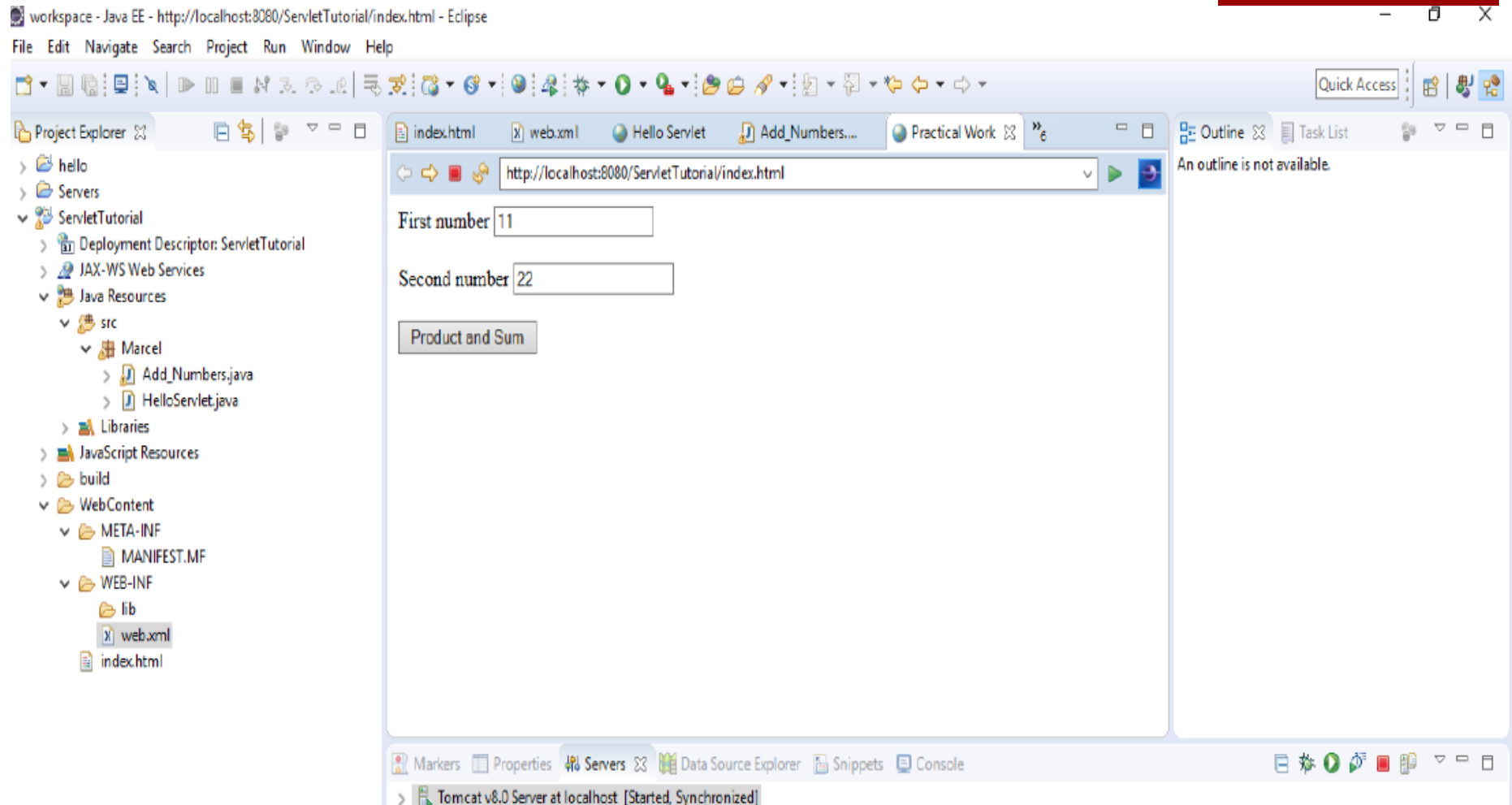
        // Get the PrintWriter object to send output back to the browser
        PrintWriter output = response.getWriter();

        // Print the sum and product results to the browser
        output.println("The Answer: " + sum + "\n\nThe Product: " + product);
    }
}
```


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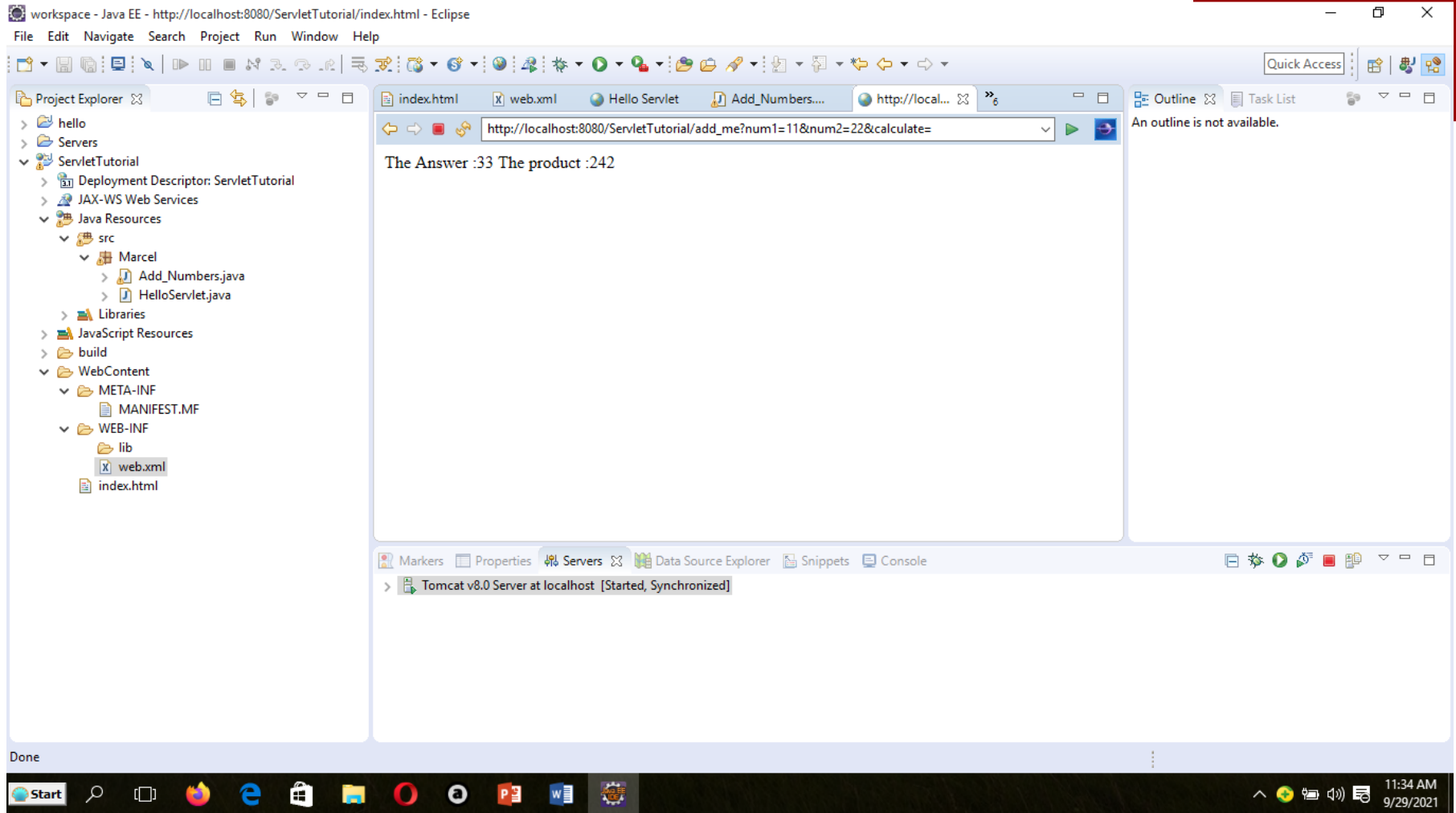
Sample Output



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Sample Output



Exercices

localhost:8080/Josbert_p1/

Sum Calculator

Calculate

Clear

✓

Sum = 28

localhost:8080/Josbbert_Servlet_sev/

Number Checker

Check if a number is positive, negative, or zero

Check Number

Clear

✓ The number is **NEGATIVE**

localhost:8080/Josbertservlet_new/

Sum Calculator

Add two numbers quickly and beautifully

45

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Calculate Clear

 Hello Dr. Josbert;
The Sum is: 79

← → ↻ ⓘ localhost:8080/p2/

Sum of Two Numbers Calculator

First Number:

Second Number:

Calculate Sum

Clear

Number Checker

localhost:8080/Josbert_Check_Number/

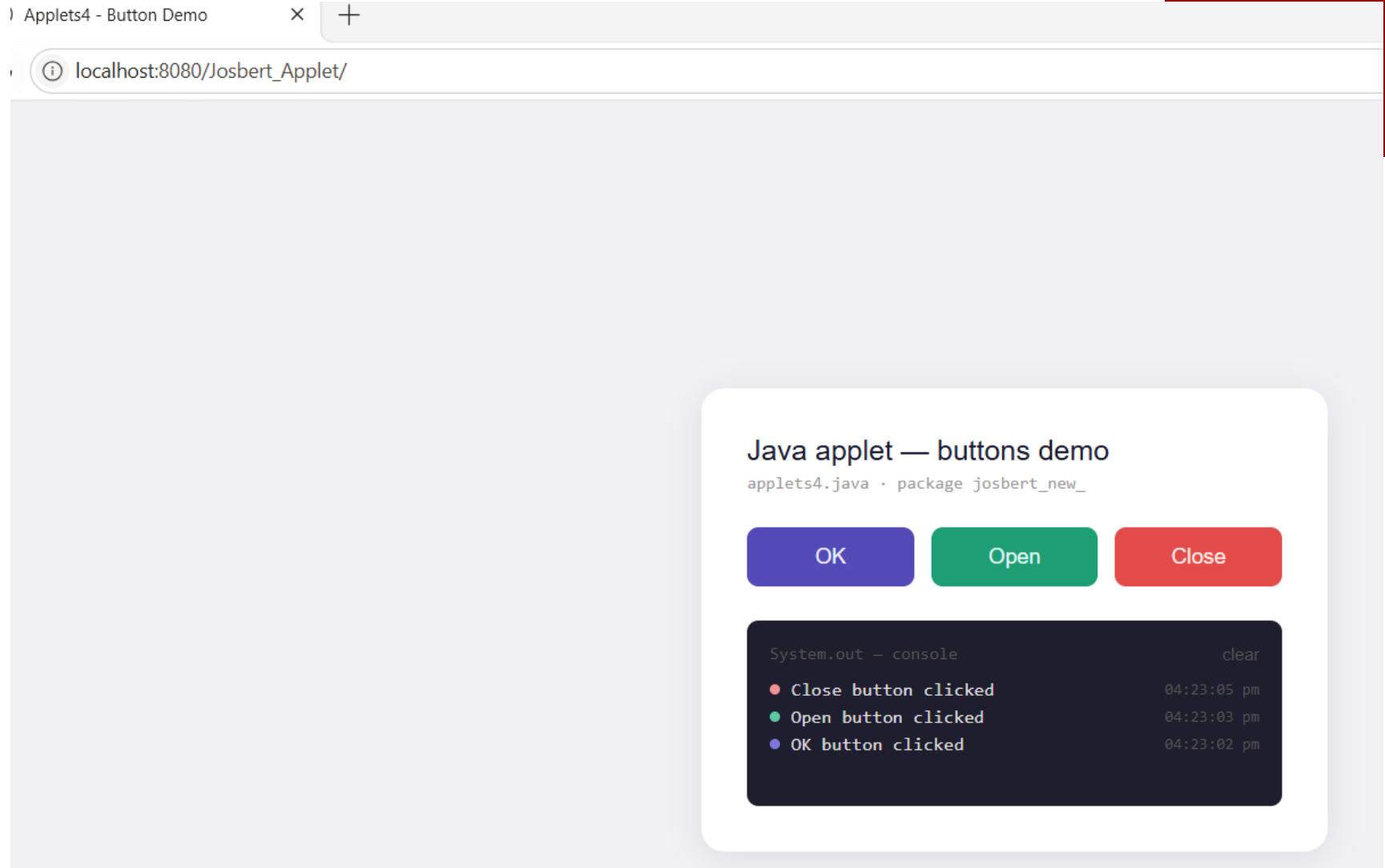
Number checker

Enter any number to check if it's positive, negative, or zero.

5

Check

 Positive number
5



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